Chapter 18 The Discrete Fourier transform



If one wishes to obtain the Fourier transform of a given function, it may happen that the function is defined in terms of a continuous independent variable, as is most often the case in books, especially in lists of transform pairs. Rut it may also happen that function values are given only at discrete values of the independent variable, as with physical measurements made at regular time intervals. Regardless of the form of the given function, if the transform is evaluated by numerical computing, the values of the transform will be available only at discrete intervals. We often think of this as though an underlying function of a continuous variable really exists and we are approximating it. From an operational viewpoint, however, it is irrelevant to talk about the existence of values other than those given and those computed (the input and output). Therefore, it is desirable to have a mathematical theory of the actual quantities manipulated.

The discrete transform formula

Questions of discreteness also arise in connection with periodic functions. A periodic function is describable by a sequence of coefficients at discrete intervals (of frequency), but this situation ***may*** be viewed as a special case of continuous frequency. The transform is then regarded as a string of equally spaced delta functions (Fig. 18.1) of strengths given by the coefficients. What if this transform were itself periodic? Then the original function would also reduce to a string of delta functions. With both function and transform now being periodic, all the information about both would be limited to two finite sets of coefficients: the strengths of the delta functions.

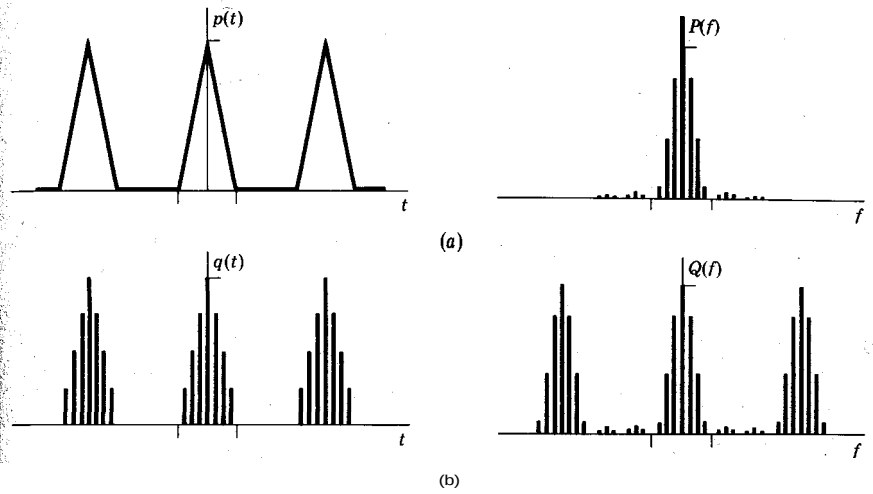


Fig. 18.1 (a) A periodic function p(t) has a transform P(f) which is a string of equispaced delta functions; (b) a periodic function q(t), which is a string of equispaced delta functions, has a transform Q(f) which is also a string of equiapaced delta functions and the string is periodic.

Thus the practical situation where a finite set of values is given and a finite set is computed does actually lie within the continuous theory. However, it pays to start afresh rather than to force the theory of the discrete Fourier transform into the continuous framework. -The reason is that the discrete notation is concise and reasonably standardized.

To retain some physical ties, let us think in terms of a signal that is a function of time; but to recognize the discreteness of the independent variable, let us use the symbol τ, which we agree can assume only a finite number N of consecutive integral values. Furthermore, we agree that τ cannot be negative. Thus, before entering into the realm of the discrete Fourier transform, we first make, if necessary, a change of scale and a change of origin. For example, suppose that a voltage waveform $v(t)$ is half a period of a cosinusoid of period 1 second (Fig. 18.2):

$$v(t) = \begin{cases} \cos 2\pi t & -0.25 < t < 0.25 \\ 0 & \text{otherwise} \end{cases}$$

and that samples are taken at intervals of 100 milliseconds. Table 18.1. shows signal values as a function of t but Table 18.2 shows how it is to be converted into a function $f(\tau)$ of discrete time τ before proceeding. This is illustrated in Fig. 18.2. In general, if the sampling interval is T and the first sample of interest occurs at $t = t_0$, then

$$f(\tau) = v(t_0 + \tau T), \qquad \tau = 0, 1, 2, \ldots, N - 1.$$

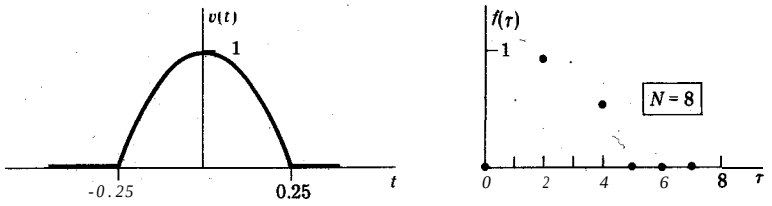


Fig. 18.2 A function of the continuous variable t and one way of representing it by eight sample values.

In what follows, $f(\tau)$ forms the point of departure. It will be noticed that no provision is made for cases where there is no starting point, as with a function such as $\exp(-t^2)$. This is in keeping with the practical character of the discrete transform, which does not contemplate data trains dating back to the indefinitely remote past. A second feature to note is that the finishing point must occur after a finite time. However, it need not come at $\tau = 5$ as in Table 18.8; one might choose to let τ run on to 15 and assign values of zero to the extra samples. This is a conscious choice that must always be made. It may be important; for example, Table 18.12 does not convey the information given in the equation preceding it-that following the half-period cosine, the voltage remains zero. The table remains silent on that point, and if it is important, the necessary number of zeros would need to be appended.

By *definition*, $f(\tau)$ possesses a discrete Fourier transform F(v) given by

$$F(\nu) = N^{-1} \sum_{\tau=0}^{N-1} f(\tau) e^{-i2\pi(\nu/N)\tau}. \qquad (1)$$

The quantity v/N is analogous to frequency measured in cycles per sampling interval. The correspondence of symbols may be summarized as follows:

	Time	*Frequency*
Continuous case	t	f
Discrete case	τ	ν/N

The symbol v has been chosen in the discrete case, instead of f, to emphasize that the frequency integer ν is *related* to frequency but is not the *same* as frequency f. For example, if the sampling interval is 1 second and there are eight samples (N = 8), then the component of frequency f will be found at $\nu = 8f$; conversely, the frequency represented by a frequency integer $\nu = 1$ will be $\frac{1}{8}$ hertz.

Table 18.1

t (milliseconds)	$v(t)$
-250	0
-150	0.588
-50	0.951
50	0.951
150	0.588
250	0

Table 18.2

τ	$f(\tau)$
0	0
1	0.588
a	0.951
3	0.951
4	0.588
5	0

Given the discrete transform F(v), one may recover the time series $f(\tau)$ with the aid of the inverse relationship

$$f(\tau) = \sum_{\nu=0}^{N-1} F(\nu) e^{i2\pi(\nu/N)\tau}. \qquad (2)$$

To see that this is so, we first verify the fact that

$$\sum_{\nu=0}^{N-1} e^{-i2\pi(\nu/N)(\tau-\tau')} = \begin{cases} N & \tau = \tau' \\ 0 & \text{otherwise} \end{cases}$$

One way is to picture the summation as a closed polygon on the complex plane, except when $\tau = \tau'$, in which case the polygon becomes a straight line composed of N unit vectors end to end. (The variable τ' is taken to assume values 0, 1, . . . , N − 1; if larger values were allowed, for example if $\tau - \tau'$ could become equal to N, $2N$, . . . , the summation would equal N for $\tau = \tau'$ mod N.) Another way to think of this is by analogy with the orthogonality relation of sinusoids of different frequencies, a viewpoint that is aided by rewriting it

$$\sum_{\tau=0}^{N-1} e^{-i2\pi(\nu/N)\tau} e^{i2\pi(\nu'/N)\tau} = \begin{cases} N & \nu = \nu' \\ 0 & \text{otherwise.} \end{cases}$$

To establish the inverse discrete transform, introduce a dummy variable τ' for convenience, and substitute (1) into the right-hand side of (2):

$$\begin{aligned} \sum_{\nu=0}^{N-1} F(\nu) e^{i2\pi(\nu/N)\tau'} &= \sum_{\nu=0}^{N-1} N^{-1} \sum_{\tau=0}^{N-1} f(\tau) e^{-i2\pi(\nu/N)\tau} e^{i2\pi(\nu/N)\tau'} \\ &= N^{-1} \sum_{\tau=0}^{N-1} f(\tau) \sum_{\nu=0}^{N-1} e^{-i2\pi(\nu/N)(\tau-\tau')} \\ &= N^{-1} \sum_{\tau=0}^{N-1} f(\tau) \times \begin{cases} N & \tau = \tau' \\ 0 & \tau \neq \tau' \end{cases} \\ &= f(\tau'). \end{aligned}$$

From the inverse transform we see that only N integral values of the frequency integer v are needed and that they range from 0 to N – 1 just as with the discrete time τ. It certainly sounds reasonable that a function defined by N measurements should be representable after transformation by just N parameters. Even when the values of f are real, the values of F are in general complex and, as will be seen below, one must be careful how to count complex numbers. The fact that v ranges over integral values, whereas the frequency ν/N is fractional, is the reason for introducing the integer v; mathematical convenience takes priority over physical significance.

In order to regain our physical feeling for numerical orders of magnitude, let us consider a record consiting of 1,024 samples separated by l-second intervals. We expect this to be representable by a Fourier series consisting of a constant term and multiples of a certain fundamental frequency. The fundamental period should be 1,024 seconds, corresponding to a fundamental frequency 1/1,024 hertz. The highest frequency needed will be 0.5 hertz, which has two samples per period. This will be the 512th harmonic. The reason that v assumes 1,024 values, whereas the number of frequencies is only 513, is as follows. If the values of f(r) are real, as is usual in records of physical quantities, there are 1,034 real data values. Now the transform $F(\nu)$ has 1,034 complex values which would require 2,048 real numbers to specify except that $F(0)$ and $F(N/2)$ have no imaginary part [see (Eq. 1)] and half the remaining values of $F(\nu)$ are complex conjugates of the other half. This is because $f(\tau)$ is real (Chapter 2). If $f(\tau)$ were complex, there would be 2,048 real data values and $F(\nu)$ would require 2,048 real numbers for its specification.

Since the highest frequency reached is 0.5 hertz, it is apparent that ν/N, which reaches a maximum value of 1,023/1,024, is not a strict analogue of frequency. For instance, where $\nu = 734$, $\nu/N = 0.707$ hertz and, as we have seen, frequencies above 0.5 hertz are neither required nor can be represented by samples at l-second intervals. Rather, v = 734 corresponds to a negative frequency –300/1,024 hertz.

This anomaly is a distinct impediment to the visualization of the connection between the Fourier transform and the discrete Fourier transform. One way to bring the two into harmony would be to redefine the discrete transform in terms of summation over negative and positive values, although it might be objected that negative subscripts are permitted only in some computer languages (for example, ALGOL). On the other hand, some widely used languages do not even allow indexing to start at *zero,* and so an index shift is required anyhow. Another objection asserts the desirability of making N an even number. Figure 18.3b

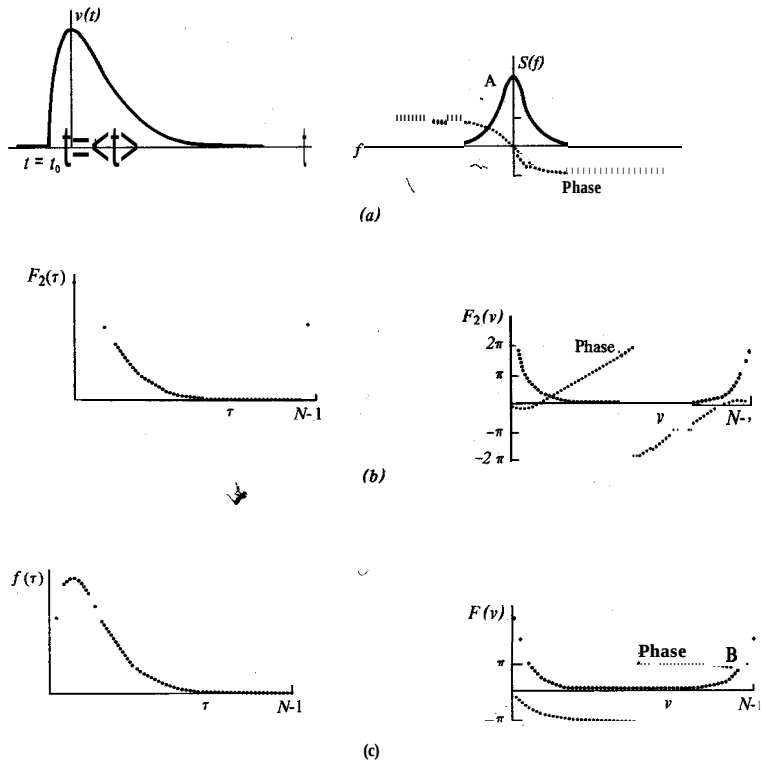


Fig. 18.3 (a) A function and its Fourier transform; (b) and (c) two way of representing the function by N samples and the corresponding discrete Fourier transforms, shown by modulus (dots) and phase (small dots).

shows how a signal $v(t)$ and its transform S(f) might be made to correspond with f(r) and its discrete transform $F(\nu)$. Alternatively, as in Fig. 18.3c, it could be arranged that $\tau = 0$ corresponds to $t = t_0$; this would not prevent the small negative values of frequency labeled A from corresponding to large positive values of v labeled B.

Finally, another way of looking at this, which is customary, is to lift the restriction of τ and ν to values from 0 to N – 1 and to allow all integral values, including negative values. The function values assigned are on the basis that $f(\tau)$ and $F(\nu)$, in their extended sense, are periodic, with period N. Thus

$$f(\tau) = f(\tau \pm N) = f(\tau \pm 2N) = \ldots$$
$$F(\nu) = F(\nu \pm N) = F(\nu \pm 2N) = \ldots .$$

This plan will be adopted here. Under this plan, the basic transform and its inverse may be written

$$F(\nu) = N^{-1} \sum_{\tau=-(1/2)N}^{(1/2)N-1} f(\tau) e^{-i2\pi(\nu/N)\tau} \tag{3}$$

$$f(\tau) = \sum_{\nu=-(1/2)N}^{(1/2)N-1} F(\nu) e^{i2\pi(\nu/N)\tau} \tag{4}$$

and ν/N may be identified with frequency measured in cycles per sampling interval over the range $-\frac{1}{2}N \leq \nu \leq \frac{1}{2}N$. If the sampling interval is T, the frequency measured in hertz is ν/NT.

In Fig. 18.4 we see a way of visualizing $f(\tau)$ and $F(\nu)$ as having cyclic dependence on τ and ν. The upper type of diagram is helpful in making a connection with our previous experience [e.g., to take the autocorrelation of a sequence, we imagine $f(\tau)$ to be rotated, in the top left-hand diagram, relative to itself]. Multiplication of corresponding values followed by summation gives a value of the autocorrelation and we can see how the result ties in with our previous knowledge that the autocorrelation of a rectangle function is a triangle function. This illuminates the lower diagrams where the original form of indexing, starting from zero, obscures the simple shapes.

Cyclic convolution

If we convolve two sequences, one having m elements and the other n, then the convolution sum, or serial product, will have m + n − 1 elements (p. 32). In particular, if we deal with sequences having N elements, then the convolution of two such sequences will not itself be containable in an N-element sequence. Figure 18.4 (top center) coped with the desire to exhibit the triangular autocorrelation of two rectangles by packing the sequence expressing the rectangle with extra zeros. Three-fourths of the elements shown at the top left are zeros, and at the top center there are still plenty of zeros left to witness to the isolated nature of the triangular island.

Suppose now that N is kept constant while the number of nonzero elements at the top left is increased. When the rectangle extends more than halfway around the circle, the outer ends of the triangle overlap. The result will be a flat, nonzero segment in the region of overlap. Clearly, the result is wrong if we are looking for the triangular autocorrelation. However, the operation exists in its own right, and may be defined as cyclic convolution.

The cyclic convolution integral $h(\theta)$ of two periodic functions $f(\cdot)$ and

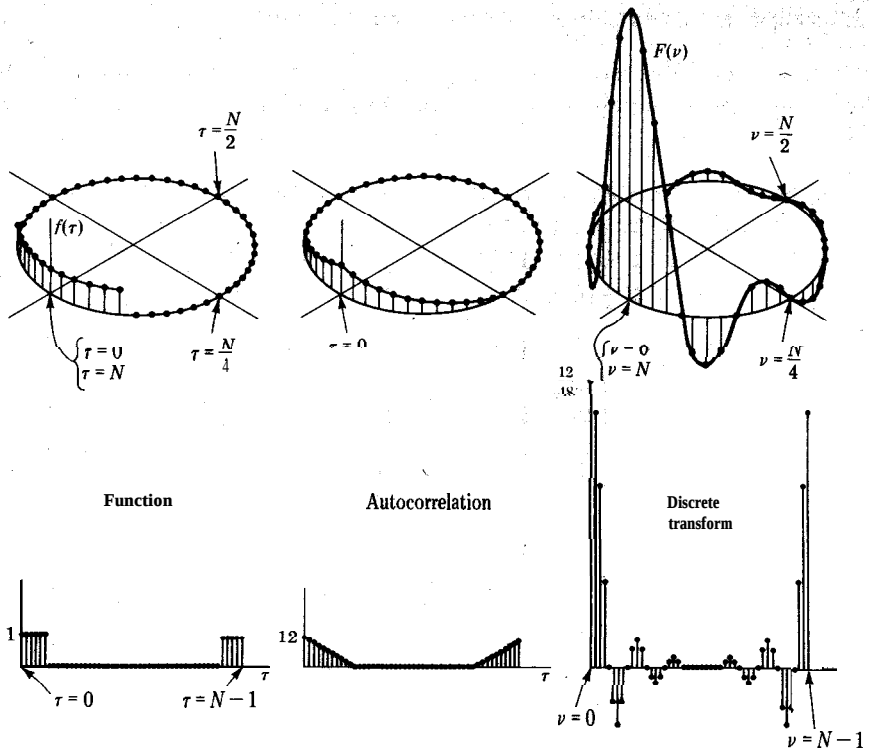


Fig. 18.4 Cyclic (above) and standard (below) representations of a sampled rectangle function, it.9 triangular autocorrelation, and its transform.

$g(\cdot)$ with period 2π is defined as

$$h(\theta) = \int_0^{2\pi} f(\theta') g(\theta - \theta')\, d\theta'.$$

The cyclic convolution sum $h(\tau)$ of two N-element sequences $f(\tau)$ and g(r) is defined as

$$h(\tau) = \sum_{\tau'=0}^{N-1} f(\tau') g[\tau - \tau' + NH(\tau' - \tau)],$$

where $H(\cdot)$ is the Heaviside unit step function. Because $\tau - \tau'$ may range from $-(N-1)$ to $N-1$ as τ and τ' range from 0 to N − 1, a term N must be added to $\tau - \tau'$ when $\tau' > \tau$ to bring $\tau - \tau'$ into the range $[0, N-1]$. This is the effect of the term $NH(\tau' - \tau)$, which will be explicitly required in computations. However, interpreting $g(\tau)$ in its extended or cyclic sense allows us to omit the term $NH(\tau' - \tau)$ from the written expressions.